

1(a)	work done per unit mass	B1
	work (done on mass) moving mass from infinity (to the point)	B1
1(b)(i)	$-3.55 \times 10^7 \text{ J kg}^{-1}$	B1
1(b)(ii)	$\phi = -\frac{GM}{r}$ $M = -\frac{-3.55 \times 10^7 \times 4800000}{6.67 \times 10^{-11}}$ $= 2.55 \times 10^{24} \text{ kg}$	B1
1(b)(iii)	$g = \frac{GM}{r^2}$ or $g = -\frac{\phi}{r}$	C1
	$= \frac{6.67 \times 10^{-11} \times 2.55 \times 10^{24}}{4800000^2}$ or $= \frac{3.55 \times 10^7}{4800000}$ $= 7.4 \text{ N kg}^{-1}$	A1
1(b)(iv)	r in range 2.60×10^7 to $2.65 \times 10^7 \text{ m}$	C1
	$\frac{mv^2}{r} = \frac{GMm}{r^2}$ and $v = \frac{2\pi r}{T}$ or $mr\omega^2 = \frac{GMm}{r^2}$ and $\omega = \frac{2\pi}{T}$	C1
	$T^2 = \frac{4\pi^2 r^3}{GM} = \frac{4\pi^2 \times (2.65 \times 10^7)^3}{6.67 \times 10^{-11} \times 2.55 \times 10^{24}} = 4.20 \times 10^9$	C1
	$T = 64\,800 \text{ s}$ $= 18 \text{ hours}$	A1

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Question	Answer	Marks
1(c)	similarity – any one point from • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance	B1
	difference – any one point from • gravitational potential is (always) negative • electric potential can be positive or negative	B1

Question	Answer	Marks
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